



- **Project Title: AI-Driven Risk Management and Decision Support for Sustainable Supply Chains**
- **Project Start Date:** 1st June 2025
- **Names of project associates/ students (if student, mentions degree and year):**
 - Puneet, Project Associate
 - Manish Kumawat
- **Problem Statement/ Description (3-4 lines):** Supply chains face unpredictable disruptions that can severely impact service levels and costs. Stochastic risk assessment enables FedEx SMART Centre to quantify these uncertainties using probabilistic models. This approach supports robust decision-making under uncertainty, aligning with FedEx's goal of building resilient, sustainable, and agile logistics solutions.

Progress against project objectives:

S. No	Objective	Progress Status
1.	Risk Assessment Framework	● Working on designing a stochastic risk assessment model integrating probability distributions of supply chain disruptions. ● Found a dataset and worked on the code to implement Monte Carlo simulations to evaluate potential outcomes of supply chain decisions under uncertainty. ● Completed Phase 1 (Literature & Concept Review), including extracting mathematical frameworks, Bayesian Network structures, and ripple-effect modeling from selected papers. ● Developed a comparison matrix of major stochastic modeling approaches. Identified and prepared publicly available datasets, conducted preliminary data analysis (EDA), and flagged data quality issues. ● Completed Monte-Carlo simulation v1 and v2, and drafted the first possible Supply Chain Risk Index (SCRI).

		● Finalized the Supply Chain Risk Index (SCRI) with calibrated severity parameters, achieving $\text{ROC-AUC} \geq 0.80$ and calibration slope within $[0.9, 1.1]$. ● Completed sensitivity and stress testing, along with distribution fitting and copula selection (Gaussian vs. Student-t). ● Incorporated calibration using EM-DAT and UNCTAD external datasets, ensuring mapping accuracy with $R^2 \geq 0.5$. ● Risk Dashboard prototype initiated and integrated with updated SCRI weights, downloadable outputs, VaR/TVaR readouts, and copula model badge. ● With these milestones, the Risk Assessment Framework is now considered functionally complete, providing a strong foundation for the next phase (AI-based decision support system). ● Enhanced the existing risk dashboard to make it fully dynamic and data-driven rather than hard-coded. Integrated updated SCRI weights and calibrated risk outputs into the dashboard views. ● Added improved visual elements and interaction logic while retaining the stable Streamlit-based UI already in use. Deployed and demonstrated the live dashboard, ensuring consistency with the calibrated Student-t copula risk model and VaR/TVaR outputs.
2.	AI-Based Decision Support System	● Built a Gym/SimPy environment with calibrated marginals and a Student-t copula sampler, ensuring seeded reproducibility and unit tests for robustness. ● Completed grid-search for (s,S) and myopic baseline policies, and established a Q-learning/DQN training harness with logged KPIs (service level, total cost, SCRI violations, VaR/TVaR). ● Introduced SCRI-penalty and CVaR-style reward terms; retrained agents and validated improvements in tail-risk metrics ($\geq 10\%$ reduction in VaR95 or $\geq 15\%$ fewer SCRI violations without significant cost increase). ● Produced Tech Report v0.5 with environment specification, baselines, QL/DQN results, ablation studies (independence vs Gaussian vs t-copula), and dashboard integration showcasing policy suggestions vs baselines. ● Significant progress was achieved toward deepening the theoretical and mathematical foundations of our reinforcement learning

		framework. We systematically built mathematical intuitions behind Q-Learning and Deep Q-Networks (DQN), analyzed convergence behavior, and explored integration mechanisms of heavy-tailed dependency models (Student-t Copulas) within the Bellman update structure. In parallel, we reviewed advanced literature on stochastic decision processes and initiated the formulation of a risk-sensitive Bellman operator that embeds copula-based correlation structures into value updates. This work lays the groundwork for developing a theoretically consistent and risk-aware AI agent for supply chain decision-making. ● Benchmarked risk-aware RL policies against classical baselines ((s, S) and myopic policies). Evaluated policies under Student-t copula-based transition dynamics to reflect heavy-tailed risk dependencies. ● Focused analysis on tail-risk KPIs (e.g., VaR95), validating measurable reduction in risk without disproportionate cost increase.
3.	Predictive Analytics for Disruptions	● Initiated and completed a first working baseline for the predictive analytics module during November. Defined the prediction target at facility level (weekly SLA-breach rate or turnaround-delay exceeding a threshold). ● Engineered features including ISO-week seasonality, holiday effects, lagged SLA/turnaround variables, and external EM-DAT/UNCTAD risk proxies. ● Implemented baseline models ranging from seasonal naïve methods to XGBoost-based tabular models. Evaluated models using regression (sMAPE) and classification (PR-AUC, F1) metrics, including calibration diagnostics.
4.	Sustainable and Resilient Logistics Strategies	Yet to do

Work completed in the project thus far:

1. Initial Research and Planning: Identified and documented publicly available datasets relevant to supply chain disruption modeling, and researched stochastic risk assessment models (Stochastic Programming, Markov Chains, Bayesian Networks, Newsvendor) with their pros, cons, and application domains.
2. Paper Review and Concept Extraction: Reviewed a key paper on "Stochastic Integrated Supplier Selection and Disruption Risk Assessment under Ripple Effect," summarizing its objective function, decision variables, constraints, uncertainty modeling, Bayesian Network structure, and ripple-effect modeling.
3. Literature Comparison: Created a matrix comparing four major stochastic modeling approaches, highlighting application domains, pros, cons, and references.
4. Seminar Participation and Summaries: Attended the FedEx/IIT Kharagpur seminar on Variance-Adjusted Cosine Similarity (VACS), summarizing key takeaways, and attended the SMART FedEx Seminar Series, summarizing insights on logistics challenges in India, global best practices, and the need for localized solutions.
5. Data Preparation (Phase 2): Reviewed the selected dataset, created a data dictionary mapping original fields to risk variables, identified data quality issues, and conducted exploratory data analysis (EDA), including histograms and ECDFs.
6. Collaboration and Repository Management: Coordinated work with team members, uploaded artifacts to a shared Google Drive and GitHub repository, and maintained logical folder structures with basic documentation.
7. Feature Engineering: Derived inter-arrival time, disruption severity proxy (Delivery_Risk x Cost), and lead-time variance, documented in feature_eng.md.
8. Data Quality Refinement: Proposed imputation/deletion rules per variable and documented them in data_quality_issues.ipynb, with updates to the README file.
9. Exploratory Statistics Polish: Added markdown interpretations for identifying heavy tails and multimodality in exploratory_statistics.ipynb, and pushed histogram and ECDF plots as PNGs to /figures/histogram_ecdfs.
10. Monte Carlo Simulation Development:
 - a. Simulated 10,000 scenarios using preliminary fitted distributions and plotted the distribution of total disruption cost (mc_simulation_v1.ipynb).
 - b. Created Monte-Carlo simulation version 2, including relevant empirical data and a separate parameters folder.
11. Copula Dependency Prototype: Developed a prototype for Gaussian copula dependency on severity and inter-arrival times, documented in copula_dependency_prototype.ipynb.
12. Supply Chain Risk Index (SCRI) Draft: Drafted the first possible SCRI and saved it in the repository as scri_draft.md.
13. Finalized SCRI: Completed the final SCRI, including an iPython notebook and CSV results. Sensitivity and Stress Testing: Performed sensitivity and stress testing analyses.
14. Calibration with External Data: Collected EM-DAT (2018–2024) and UNCTAD port metrics, aggregated to weekly granularity, and mapped disruption severity proxies. Built calibration.ipynb and validated mapping with $R^2 \geq 0.5$.
15. Copula Selection and Diagnostics: Compared Gaussian vs. Student-t copulas on calibrated marginals. Reported AIC/BIC, log-likelihood, Kendall's τ , and tail-dependence λ . Selected the Student-t copula based on fit and tail exceedance accuracy.
16. SCRI Re-weighting and Validation: Re-weighted the Supply Chain Risk Index (SCRI) with calibrated severity parameters. Recomputed node-level scores, achieving ROC-AUC ≥ 0.80 and calibration slope within $[0.9, 1.1]$, using Platt scaling and isotonic regression.
17. Dashboard Enhancements: Updated the Streamlit risk dashboard with model badge (e.g., "t-copula"), downloadable SCRI results, VaR/TVaR readouts, and a short "What changed after calibration" note.
18. MDP Specification & KPIs: Finalized Markov Decision Process (MDP) formulation with defined KPIs including service level, total cost, SCRI-violation count, and VaR/TVaR at weekly horizons. Documented and pushed to GitHub.

19. **Simulation Environment Development:** Built a Gym/SimPy environment using calibrated marginals with Student-t copula sampling. Added seeded reproducibility and unit tests to ensure deterministic behavior across experiments.
20. **Baseline Policy Implementation:** Implemented and benchmarked (s,S) and myopic baseline policies. Conducted grid-search runs, logged KPIs, and produced comparative plots and CSV reports for baseline vs. Q-learning performance.
21. **Learning Agent Prototyping:** Developed initial Q-learning and DQN harnesses, including experience replay, target networks, and logging of learning curves. Saved policies and configs under structured artifact directories for reproducibility.
22. **Risk-Aware Reinforcement Learning:** Extended agent reward functions with SCRI-penalty and CVaR-style terms. Performed ablation studies (independence vs. Gaussian vs. t-copula; severity $\pm 20\%$, inter-arrival variability $\pm 15\%$). Validated that risk-aware agents improved tail-risk metrics ($\geq 10\%$ reduction in VaR95 or $\geq 15\%$ fewer SCRI-violations) without $> 5\%$ cost escalation.
23. **Reinforcement Learning Foundations and Mathematical Deep Dive:** Conducted a comprehensive study to strengthen the mathematical underpinnings of Q-Learning and Deep Q-Networks (DQN). Derived the Bellman expectation and optimality equations from first principles, examined convergence conditions under stochastic reward structures, and mapped how parameterized Q-functions approximate the true value function. Documented learning dynamics (exploration–exploitation, discount factor effects, target update stability) and created a unified reference for all reinforcement learning agents developed in the project. This mathematical deep dive has enhanced the theoretical consistency and interpretability of our implemented agents.
24. **Copula Integration Research and Student-t Formulation for Risk-Aware Bellman Updates:** Explored and mathematically formulated how dependency structures captured by Student-t Copulas can be integrated into the Bellman update equation. Studied joint tail dependencies between cost and disruption variables, derived conditional expectation expressions, and identified potential modifications to the reward and transition components to embed heavy-tailed correlations. This investigation establishes the theoretical pathway toward a copula-augmented, risk-sensitive Q-learning framework, enabling more robust and realistic decision policies under correlated uncertainty.
25. **Predictive Analytics Baseline Development:** Developed the first working baseline for the predictive analytics module by defining a facility-level prediction target (weekly SLA-breach rate or turnaround-delay exceedance). Engineered features including ISO-week seasonality, holidays, lagged operational indicators, and external EM-DAT/UNCTAD disruption proxies. Implemented baseline models ranging from seasonal naïve forecasting to XGBoost-based tabular models. Evaluated performance using regression (sMAPE) and classification (PR-AUC, F1) metrics, including calibration analysis. Delivered complete, reproducible artifacts—baseline notebook, metrics CSV, and forecast-versus-actual visualization—to the project repository.
26. **Dashboard Dynamization and Integration:** Converted the risk dashboard from partially hard-coded logic to a fully dynamic, data-driven architecture. Integrated predictive outputs, calibrated SCRI measures, and updated risk metrics into the dashboard framework. Ensured alignment with the Student-t copula-based risk model and consistent VaR/TVaR reporting. Deployed and demonstrated the live dashboard with improved interactivity, maintainability, and end-to-end integration across risk assessment, prediction, and decision-support components.

Specific work done this month:

- Built and validated a predictive analytics baseline for facility-level disruption forecasting, covering target definition, feature engineering, model training, evaluation, and calibration.
- Engineered temporal, operational, and external risk-proxy features (ISO-week seasonality, holidays, lagged indicators, EM-DAT/UNCTAD proxies) for predictive modeling.
- Implemented and evaluated baseline models ranging from seasonal naïve approaches to XGBoost, reporting sMAPE, PR-AUC, F1, and calibration diagnostics.
- Delivered complete predictive artifacts to the repository, including executable notebooks, metrics files, and forecast-versus-actual visualizations.
- Strengthened the risk-aware reinforcement learning setup by benchmarking RL policies against classical (s, S) and myopic baselines under Student-t copula dynamics.
- Analyzed RL performance with emphasis on tail-risk KPIs (VaR95, SCRI violations) and cost–risk trade-offs.
- Enhanced the theoretical grounding of the RL framework by aligning empirical behavior with Bellman updates under stochastic and correlated uncertainty.
- Converted the risk dashboard from partially hard-coded logic to a fully dynamic, data-driven architecture.
- Integrated calibrated SCRI outputs, predictive results, and VaR/TVaR metrics into the live dashboard.
- Deployed and demonstrated the updated dashboard with improved interactivity and consistency across risk assessment, prediction, and decision-support modules.

Project Outcome for the month:

- Successfully transitioned the predictive analytics module from planning to a working, evaluated baseline, demonstrating feasibility for facility-level disruption forecasting.
- Established end-to-end integration between risk assessment, predictive analytics, and decision-support components through a unified, dynamic dashboard.
- Demonstrated measurable improvements in tail-risk behavior through risk-aware reinforcement learning policies when compared against classical baselines.
- Delivered a stable, reproducible technical stack (models, simulations, dashboards) suitable for demonstration and further extension once additional FedEx data becomes available.
- Positioned the project to move from baseline development to deeper modeling, validation, and deployment-oriented enhancements in subsequent phases.

Next Steps (Work Plan for next month):

- Execute a focused pilot phase to demonstrate that risk-aware reinforcement learning policies outperform strong classical baselines on simulated supply-chain lanes using calibrated marginals and Student-t copula dependencies.
- Finalize a lightweight MDP formulation, including state, action, transition, and risk-aware reward definitions incorporating SCRI and optional CVaR-based tail penalties.
- Define and run targeted simulation scenarios (normal conditions, high-severity stress, and clustered disruptions) with multiple random seeds to ensure robustness.
- Implement and grid-search classical baseline policies ((s, S) and myopic) and benchmark their performance across defined scenarios.
- Train and evaluate tabular Q-learning and lightweight DQN agents under identical simulation settings and acceptance criteria.
- Assess policy performance using primary (weekly VaR95 of cost) and secondary (mean cost, service level, SCRI violations) metrics.
- Accept the pilot as successful if risk-aware agents achieve at least a 10% reduction in VaR95 with no more than a 5% increase in mean cost relative to the best baseline.
- Organize code, configurations, experimental outputs, and learned policies in a reproducible repository structure with fixed seeds and logged configurations.
- Extend the dashboard with a policy comparison tab showing baseline versus agent recommendations along with VaR and mean-cost readouts.
- Prepare a concise pilot summary documenting methods, scenarios, metrics, results, and identified gaps to guide subsequent iterations or deployment once FedEx data becomes available.

Project Progress:**Project Timeline and current work activity:**

Phase/Work Tasks	Sub-Tasks	June	July	August	September	October
3.1 Risk Assessment Framework	Literature review and model design	Completed				
	Paper extraction & modeling approach	Completed				
	Data sourcing and initial mapping	Completed				
	Data cleaning and quality assessment		Completed			
	Exploratory data analysis and distribution fitting		Completed			

	Monte Carlo simulation prototype		Completed			
	Multi-factor risk index design			Completed		
	Risk index testing and refinement		Completed			
	Distribution Fitting & Goodness-of-Fit (GoF) Matrix		Completed			
	Alternative Copula Check			Completed		
	Sensitivity and Stress Testing			Completed		
	Risk Dashboard Prototype			Completed		
	Deployment and DevOps			Completed		
AI-Based Decision Support System	Environment and Baseline Setup				Completed	
	Baseline Implementation and Training Harness				Completed	
	Risk Aware Learning and Analysis				Completed	
	Deep Dive into RL Agents					Start
	Mathematical Study of Student t Copula					Start
	FedEx terminology research for dataset preparation					Start
	Packaging and Demo Preparation					Start

Work Contributions by individual team members in the project:

1. Puneet

- Independently designed and implemented the predictive analytics baseline, including target formulation, feature engineering, model development, evaluation, and calibration.
- Developed and validated baseline forecasting and machine learning models, and delivered all predictive artifacts to the project repository.

- c. Strengthened the risk-aware reinforcement learning framework through benchmarking, tail-risk analysis, and cost–risk trade-off evaluation.
- d. Conducted theoretical deep-dives to reinforce the mathematical foundations of Q-learning, DQN, and risk-aware Bellman updates under correlated uncertainty.
- e. Converted the existing risk dashboard into a fully dynamic, data-driven system and integrated predictive and calibrated risk outputs.
- f. Deployed, tested, and demonstrated the updated dashboard as a unified interface for risk assessment, prediction, and decision support.